

MCS-1: Probability and Statistics (MA6.302)

Monsoon 2026

Lecture 11

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Bayes Law with Continuous r.v. but with Discrete hidden Cause

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- Pretend Y is a discrete r.v. Then,

$$P(A|Y = y) = \frac{P(Y = y|A)P(A)}{P(Y = y|A)P(A) + P(Y = y|A^c)P(A^c)}$$

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- All that changes for a continuous r.v. is:

$$P(A|Y = y) = \frac{f_{Y|A}(y)P(A)}{f_{Y|A}(y)P(A) + f_{Y|A^c}(y)P(A^c)}$$

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- With flu, Y is a normal random variable with $\mu = 102$ and $\sigma = 2$.

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- With flu, Y is a normal random variable with $\mu = 102$ and $\sigma = 2$.
- My temperature is 100. If A is the event “has flu” and Y is temp.

$$f_{Y|A}(y) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y-102)^2}{2 \times 4}}$$

$$f_{Y|A^c}(y) = \frac{1}{0.5\sqrt{2\pi}} e^{-\frac{(y-98.6)^2}{2 \times 0.25}}$$

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- What will be $P(A|Y = y)$?

$$P(A|Y = y) = \frac{f_{Y|A}(y)P(A)}{f_{Y|A}(y)P(A) + f_{Y|A^c}(y)P(A^c)}$$

$$P(A|Y = 100) = \frac{0.2 \times \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y-102)^2}{2 \times 4}}}{0.2 \times \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y-102)^2}{2 \times 4}} + 0.8 \times \frac{1}{0.5\sqrt{2\pi}} e^{-\frac{(y-98.6)^2}{2 \times 0.25}}}$$

Bayes Rule with continuous r.v. and continuous hidden cause

- What is $f_{X|Y}(x|y)$?

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{\int_{-\infty}^{\infty} f_{Y|X}(y|x)f_X(x)dx}$$

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- How about the conditional expectation of some function $g(X)$ given some event A ?
- $\mathbb{E}[g(X)|A] = \int_{-\infty}^{\infty} g(x)f_{X|A}(x)dx$

Total Expectation Theorem

- More generally, if $A_1, A_2, \dots, A_n$ are a partition of Ω , we have a continuous version of the total expectation theorem:

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- Or, if we are conditioning on specific values $Y = y$,
$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \mathbb{E}[X|Y = y] f_Y(y) dy$$

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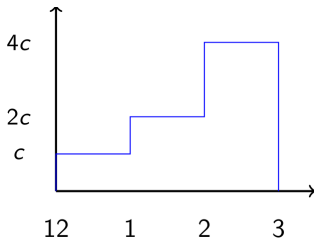
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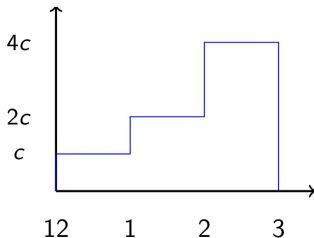
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- $P(X > 2) = \int_2^3 f_X(x) dx = 4/7$
- Similarly, $P(X < 1) = \int_0^1 f_X(x) dx = 1/7$ and $P(1 \leq X \leq 2) = 2/7$.

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- So, we can write the total expectation as

$$\begin{aligned}\mathbb{E}[X] &= \int_0^1 xP(X < 1)f_{X|X<1}(x)dx + \int_1^2 xP(1 \leq X \leq 2)f_{X|1 \leq X \leq 2}(x)dx \\ &\quad + \int_2^3 xP(X > 2)f_{X|X>2}(x)dx \\ &= \mathbb{E}[X|X < 1]P(X < 1) + \mathbb{E}[X|1 \leq X \leq 2]P(1 \leq X \leq 2) \\ &\quad + \mathbb{E}[X|X > 2]P(X > 2) \\ &= 0.5 \times 1/7 + 1.5 \times 2/7 + 2.5 \times 4/7 = 27/14\end{aligned}$$